A-A-53384A
19 June 1989
SUPERSEDING
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14 March 1988

COMMERCIAL ITEM DESCRIPTION

CEMENT, ZINC OXIDE AND EUGENOL, DENTAL (Encapsulated)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers zinc oxide and eugenol cement in pre-loaded capsules used as an intermediate restorative material.

Salient Characteristics:

The cement shall consist of powder and liquid components in pre-loaded capsules. The capsules shall contain the powder and liquid components in separate compartments without leakage or degradation. The capsules shall be activated by twisting and tightening the top of the capsule. The cement shall triturate into a smooth and creamy mixture in 12 seconds or less with an electric amalgamator, variable speed, 2900 to 4300 rpm.

Material. The cement shall consist of ivory colored zinc oxide powder and eugenol liquid in a ratio within the range from 5:1 up to 6:1 by weight. Each capsule shall contain a total of 0.34 g of cement (powder and liquid). The powder shall consist of zinc oxide, polymethyl methacrylate and zinc acetate and shall contain no quartz or alumina abrasive filters. The liquid shall consist of eugenol and acetic acid. The eugenol and zinc acetate components shall be in accordance with the tests, standards and requirements of the current Pharmacopeia of the United States of America. Zinc oxide shall be in accordance with the tests, standards and requirements of the current Pharmacopeia of the United States of America for lead and arsenic content only.

The acetic acid shall conform to the requirements of analytical reagent chemicals of American Chemical Society (A.C.S.) grade contained in the American Chemical Society publication, "Reagant Chemicals, A.C.S. Specifications".

AMSC N/A FSC 6520

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Consistency. The material, when mixed, shall be smooth and creamy with a consistency of 16.5 mm to 19.5 mm when tested as specified in test section under consistency.

Film thickness. The film thickness of the material shall be not less than 50 microns and not more than 90 microns when tested as specified in test section under film thickness.

Compressive strength. The compressive strength shall be 9000 pounds per square inch, minimum, when tested as specified in test section under compressive strength.

Setting time. The setting time shall be 8.5 minutes to 12.0 minutes when tested as specified in test section under setting time.

Solubility. The solubility of the cement shall be no greater than 0.2 percent by weight when tested as specified in test section under solubility.

Instructions. Accurate and adequate instructions for using the cement shall accompany each unit package. The instructions shall include mixing methods, mixing time and setting time.

Stability. The shelf life of the cement shall be no less than 24 months from date of manufacture. The inspection test date shall be 24 months from date of manufacture. Not more than 4 months shall have elapsed from the date of manufacture of the cement to the date of delivery to the Government.

Workmanship. The material shall be free from dirt, foreign materials and impurities which detract from its appearance or impair its serviceability.

Unit. Package (PG). One package containing two hundred capsules of zinc oxide and eugenol cement with instructions for use, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of cement capsules. Unless otherwise specified, sampling for examination shall be conducted at inspection level S-2, AQL 2.5 (percent defective).

Examination. The cement capsules shall be examined for defects, including, but not limited to, those listed below.

Defects.

Damaged package such as broken capsules, leaks, etc.
Top of capsule not properly prepared such as missing tops, tops already tightened, etc.
Instructions missing, incorrect, or incomplete.
Material not free from foreign material.
Color not as specified.

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Tests. Tests shall be conducted to determine compliance with specification requirements. The unit of product for testing shall be one package of dental cement. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Preparation of test specimens. The preparation of test specimens shall be conducted at a temperature of 21 to 24 degrees Centigrade (^{o}C), and 45 to 50 percent relative humidity. A sufficient number of capsules shall be used to obtain samples required for each test.

Consistency. The apparatus used for measuring consistency consists of two flat glass plates, a weight, a gauge plug, and a glass tube (inside diameter approximately 6.5 millimeters (mm), which will deliver 0.50 ml of mixed cement. The combined weight of the top plate and the weight shall be 120 g. First, 0.50 $\pm$ 0.02 ml of sample shall be delivered from the glass tube onto a flat glass plate. Three minutes after the mixing is started, the top glass plate weighing approximately 20 g, and the necessary additional weight required to total 120 g shall be carefully lowered onto the soft cement. Ten minutes after starting the mix, the average of the major and minor diameters of the slumped mass of cement shall be the test consistency of the mixed cement. The average of three such determinations shall be the size required to produce a mix of standard consistency.

Film thickness. A portion of mix of standard consistency shall be placed between two flat square or round glass plates of uniform thickness. The surface areas of the plates between which the cement is spread shall be approximately 2 square centimeters. Three minutes after the mix is started, a load of 15 kilograms shall be applied vertically on the top plate. Ten minutes after the mix is started, the thickness of the two plates with the cement film between them shall be determined. The difference in the thickness of the plates with and without the cement film shall be considered as the film thickness. An average of three tests shall be reported to the nearest 5 microns.

Compressive strength. Five test specimens shall be prepared. The test specimens shall be cylinders 12 mm high and 6 mm in diameter. The ends of the specimens shall be flat and smooth and shall be parallel to each other and at right angles to the long axis of the cylinder. A cylindrical mold 12 mm high and 6 mm in diameter shall be placed on a flat glass plate and slightly overfilled with cement of standard consistency within three minutes after commencing the mixing. A second flat glass plate shall be pressed on top of the mold. The mold and plates shall be held firmly together with a small "C" clamp. All apparatus shall be at room temperature.

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The molds shall be made of hard rubber, glass, stainless steel or any other substance which will not react with the cement. The molds may be coated with a 3 percent solution of a microcrystalline wax (melting point 86 to 91 ^{o}C) in benzene. Three minutes after starting the mix, the mold and clamp shall be transferred to an atmosphere of 100 percent relative humidity at 37° C. One hour later the ends of the cylinder shall be surfaced plane at right angles to the axis. The ends of the specimens may be ground flat by the use of a small amount of 200-mesh silicon carbide powder and water. The molds containing the specimens shall be drawn back and forth across a glass plate coated with the abrasive and water. They shall be rotated about one-fourth turn every few strokes. The test specimens shall be kept wet during the grinding. A small piece of cloth, wet with distilled water, shall be wrapped around circumference of the specimen. After surfacing, the specimens may be removed from the mold by a screw jack and shall be stored for one week in water at 37° C before crushing. Test equipment used for determining compressive strength shall be the Instron Engineering Corporation's Test Machine, or other suitable equipment, provided prior approval is received from the contracting officer. Testing conditions on the Instron test machine shall be as follows:

a. Crosshead speed of 0.2 in. per minute.
b. Force range of 1000 pounds.
c. Chart speed of 10 in. per minute.

The value for compressive strength shall be reported as the average of three or more from a lot of five specimens and shall be rounded off to the nearest 50 pounds. If the values for individual specimens fall more than 15 percent below the average of the five, they shall be discarded and the average of the remaining specimens shall be reported. In case more than two of the specimens are eliminated, the test shall be repeated.

Setting time. A mix of cement shall be loaded into a polyethylene crown form cap with approximate internal dimensions of 9.5 mm diameter by 7 mm deep. Type A durometer hardness shall be determined as described in American Society for Testing and Materials (ASTM) Designation D2240. Readings shall be taken at 0.5 minute intervals. Hardness shall be plotted against time, and the setting time in minutes shall be the interpolated value of time when durometer reading equals 50. Setting time shall be reported to the nearest 0.5 minute.

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Solubility. One half milliliter of mixed cement shall be placed in a split stainless steel ring (inside diameter 20 mm and 1.5 mm thick) which is sitting on a thin polyethylene sheet backed by a flat plate. Another flat plate faced with a sheet of thin polyethylene shall be used to press the cement into the ring. A piece of fine platinum or corrosion-resistant wire shall be placed in the soft cement as the specimens are formed to provide a convenient means of holding the specimens. Three minutes after the mix is started, the plates and cement shall be placed for one hour in an atmosphere having a relative humidity of 100 percent at 37° C. After one hour, the specimens shall be withdrawn from the oven and two specimens shall be placed in each of two tared weighing bottles and weighed. The combined weight of the two specimens and the weighing bottle, less the weight of weighing bottle and the platinum wire, shall be taken as the weight of the specimens of cement. The specimens shall immediately be submerged by pouring 50 ml of distilled water into the weighing bottle, which shall be stored for 23 hours at 37° C. The specimens shall then be removed from the water. There shall be no evidence of crystal growth on the surface of the specimens. The water shall be evaporated from the weighing bottle at a temperature just below 100° C (212° F). The weighing bottle shall then be dried at 150° C (302° F) to constant weight. After cooling to room temperature in a desiccator containing thoroughly dry anhydrous calcium sulfate ($CaSO_4$) or silica gel (freshly dried at 130° C), the weighing bottle and contents shall be weighed with a precision of 0.2 milligram (mg). This cycle of heating the weighing bottle to 150° C, cooling over a desiccant and reweighing shall be repeated until the weight loss of each bottle is no more than 0.5 mg in any 24-hour period. The difference between the final weight of the weighing bottle and its initial weight shall be the amount of disintegration. The gain in weight divided by the weight of the specimens times 100 gives the percent of disintegration. The average of duplicate tests (two weighing bottles containing two specimens each) shall be reported to the nearest 0.1 percent.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: 1. The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-01-263-6284	CEMENT, ZINC, OXIDE and Eugenol, Capsules, 200s.

NOTE: 2. The following National Stock Number is referenced in this document.

National Stock Number	Item Identification
6520-00-149-0123	AMALGAMATOR, ELECTRIC, DENTAL Variable Speed, 115 volts, 60Hz, AC.

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Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2231

LOCATION: NCR DISK # 5

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